

CHAPTER 2

Introducing economic primitives

The strength of the Anthropic Economic Index lies in showing not only how much AI is used, but how it is used. In prior reports, we showed which tasks Claude is used for, and how people collaborate with Claude. These data have enabled external researchers to analyze labor market shifts (e.g., [Brynjolfsson, Chandar & Chen, 2025](#)).

In this edition of the Anthropic Economic Index, we expand the breadth of data available to external researchers by providing insights on five economic “primitives”, by which we mean simple, foundational measures of the ways that Claude is used, which we generate by asking Claude to answer specific questions about the anonymized transcripts in our sample. Some of our primitives encompass several such questions, and others use a single indicator.

Because AI capabilities are advancing so rapidly and the economic effects will be unevenly experienced, we need a breadth of signals to uncover not just how Claude is used but also to inform what impact this technology will have.

Dimensions of AI use that matter for economic impacts

This report introduces five new economic primitives beyond the one we already measure, collaboration patterns (whether users automate or augment their tasks with Claude). These primitives capture five dimensions of a human-AI conversation: 1) task complexity, 2) human and AI skills, 3) work, coursework or personal use case, 4) the AI’s level of autonomy, and 5) task success (see Table 2.1). AI autonomy captures something different from our existing automation/augmentation distinction. For example, “Translate this paragraph into French” is high automation (directive, minimal back-and-forth) but low AI autonomy (the task requires little decision-making from Claude).

Task complexity captures that tasks can vary in their complexity, including how long they take to complete and how difficult they are. A “debugging” task in O*NET could refer to Claude fixing a small error in a function or comprehensively refactoring a codebase—with very different implications for labor demand. We measure complexity through estimated human time to complete tasks without AI, time spent completing tasks with AI, and whether users handle multiple tasks within a single conversation.

Human and AI skills address how automation interacts with skill levels. If AI disproportionately substitutes for tasks requiring less expertise while complementing higher-skilled work, it could be another form of skill-biased technical change—increasing demand for highly skilled workers while displacing lower skilled workers. We measure whether users could have completed tasks without Claude, and the years of education needed to understand both user prompts and Claude’s responses.

Use case distinguishes professional, educational, and personal use. Labor market effects most directly follow from workplace use, while educational use may signal where the future workforce is building AI-complementary skills.

AI autonomy measures the degree to which users delegate decision-making to Claude. Our latest report documented rising “directive” use where users delegate tasks entirely. Tracking autonomy levels—from active collaboration to full delegation—helps forecast the pace of automation.

Task success measures Claude’s assessment of whether Claude completes tasks successfully. Task success helps assess whether tasks can be automated effectively (can a task be automated at all?) and efficiently (how many attempts would it take to automate a task?). That is, task success matters for both the feasibility and the cost of automation labor tasks.

Selecting and validating the new measures

The new dimensions of AI use captured in our data were informed by our recent work on the [productivity effects of Claude](#), feedback we received from external researchers, recent literature on AI’s economic impact through the lens of human capital and expertise ([Vendraminell et al., 2025](#)), and